EMODnet Marine Knowledge for Non-Governmental Organisations Online workshop, 23 September 2025

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Executive Summary

On 23 September 2025, the European Marine Observation and Data Network (EMODnet)[1], in collaboration with the European Commission's Directorate-General for Maritime Affairs and Fisheries (DG MARE), co-organised an online workshop titled "EMODnet Marine Knowledge for Non-Governmental Organisations (NGOs)." The workshop aimed to foster dialogue between EMODnet and NGOs active in the ocean domain. Its objectives were to raise awareness of EMODnet and its extensive portfolio of marine data derived from *in situ* (field-based) ocean observations, exchange insights on how NGOs currently use, or could further leverage, EMODnet as a foundational source of marine knowledge, gather feedback on existing and emerging marine data needs of NGOs in the context of the European Ocean Pact, and explore opportunities for data sharing with EMODnet while discussing how such collaboration could be scaled within the NGO sector.

The workshop was structured around two plenary sessions focused on data use and data sharing, guided by targeted discussion questions to facilitate open dialogue. This event took place shortly after the launch of the European Ocean Pact, which serves as a unified reference framework for EU ocean-related policies. Successful implementation of these policies rely on open access to high-quality marine knowledge to support implementation efforts, including the EU Marine Strategy Framework Directive (MSFD) and the goal of achieving Good Environmental Status of European seas by 2030, and the Nature Restoration Law, among others, and EU obligations to global treaties including Biodiversity Beyond National Jurisdiction and broader policy frameworks and conservation targets, such as the 30x30 initiative to protect 30% of the world's ocean by 2030.

This workshop forms part of EMODnet's ongoing stakeholder engagement efforts, complementing previous dialogues with Blue Economy sectors (2020–2024), including aquaculture, offshore renewable energy, coastal tourism, and recreational boating, marinas and ports.

Welcome and Opening Remarks

Kate Larkin (KL, Head of EMODnet Secretariat) opened the workshop as Chair, welcoming all participants and providing housekeeping information and an overview of the agenda and main objectives of the meeting. She noted that the workshop brought together a diverse network of Blue NGOs working on ocean-related issues, including those with Brussels-based offices operating at European and global levels, as well as NGOs with established connections to EMODnet who could share examples of engagement at national, regional, and local levels, based on a stakeholder mapping conducted by the Secretariat, with input from EMODnet Coordinators. She welcomed all NGO participants, representatives from the European Commission Directorate-General for Maritime Affairs and Fisheries (EC DG MARE), EMODnet thematic and data ingestion coordinators and wider experts.

Zoi Konstantinou (ZK, EC DG MARE) delivered the opening address, setting the scene within the broader EU policy context. She stressed the growing significance of marine knowledge in addressing global environmental challenges and adapting to shifting policy priorities and noted the value of the EU's flagship marine data services EMODnet and Copernicus Marine, and their collaboration for the EDITO long-term infrastructure of the European Digital Twin Ocean, noting that these three public marine knowledge assets exemplify the EU's commitment to environmental stewardship.

ZK spoke about the important work of NGOs, highlighting the need of NGOs for accessible, high-quality marine data for their diverse activities, and noting that NGOs are increasingly also collecting their own data. She underlined that EMODnet stands as a cornerstone of marine knowledge, providing

[1] https://emodnet.ec.europa.eu/en

standardized, authoritative datasets on the marine environment and human activities at sea that support NGO activities. She expressed a strong interest in EMODnet to foster structured collaboration with NGOs, including the integration of citizen science, and emphasized the collective value of NGO contributions to enhance marine knowledge services and products.

Following the opening remarks, a tour de table was then conducted.

Session 1: EMODnet for NGOs: The offer, uses and applications

1.1 EMODnet Overview

KL opened the first session by presenting an overview of EMODnet, stressing that effective implementation of ocean and environmental policies—such as the European Ocean Pact—depends on integrated, multi-parameter, and interoperable marine knowledge covering ocean, coastal zones, and the land-sea interface.

She highlighted that EMODnet is the European Union's *in situ*[2] marine data service, managed by EC DG MARE. She underlined that EMODnet provides a public, open-access platform for marine and coastal environmental and human activities data, with the diverse offer of 68 parameters providing actionable data that supports the implementation of numerous EU Directives, e.g., Marine Strategy Framework Directive (MSFD), Maritime Spatial Planning Directive (MSPD) and international policy frameworks. And she highlighted that marine knowledge was a cross-cutting enabler for the European Ocean Pact[3] and that the demand for EMODnet was continuing to increase and diversify to provide the foundational marine data to support the Ocean Pact implementation, which includes the EU Ocean Observation Initiative and the European Digital Twin Ocean. KL explained EMODnet's value chain, which begins with *in situ* ocean observations with EMODnet assembling disparate data collection efforts and conducting data curation, applying EU and international data and metadata standards, and creating data products that are fully open and free to use, harmonised at the European scale and beyond. The EMODnet offer covers local, regional sea-basin and European scale, with some EMODnet thematics offering data from beyond European waters, such as the Caribbean Sea, with some EMODnet thematics providing a global offer for some parameters. KL highlighted the EMODnet Common Map Viewer[4] (Figure 1), as the single point of access for discovering, visualising, and downloading all EMODnet data and data products. She added that this is underpinned by an extensive central metadata catalogue and can be accessed via multiple data and web services.

KL highlighted the critical role of *in situ* data for delivering high-resolution spatial and temporal insights and that the value of *in situ* goes well beyond its use in validating satellite observations, and as input to models, with *in situ* offering integrated multi-parametric data which are essential at the local to regional scale. She outlined EMODnet's collaborations with diverse stakeholder groups that are data providers and users, spanning research and academia to national administrations and policy, the private sector, including the Blue Economy, non-governmental organisations and society. She highlighted that EMODnet is already receiving data from citizen science initiatives, which help fill data gaps in long-term established ocean observing programmes, and help promote ocean and data literacy. KL underlined EMODnet's key role in the European Digital Twin of the Ocean, where EMODnet

[2] *In situ* ocean observations are collected by sensors and samplers placed in seawater, on/in the seafloor, the coastal zone and surrounding air (see communication on the *in situ* marine data landscape in Europe : https://emodnet.ec.europa.eu/sites/emodnet.ec.europa.eu/files/public/PDF/EUInSituMarine_EMODnet_CMEMS_FINAL.pdf

[3] https://oceans-and-fisheries.ec.europa.eu/european-ocean-pact_en

[4] https://emodnet.ec.europa.eu/geoviewer/

provides the *in situ* data component for next-generation, cloud-based decision-support tools for society and the EU, working in direct collaboration with Copernicus Marine Service to deliver the EDITO long-term infrastructure of the European Digital Twin Ocean.

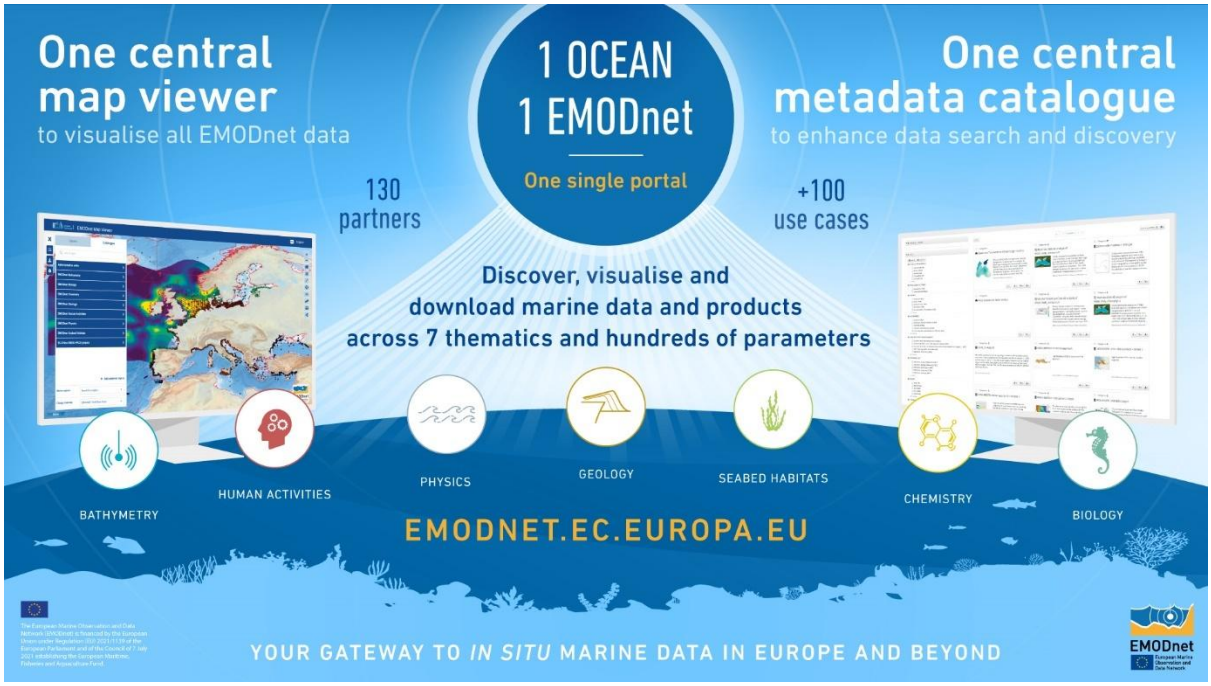


Figure 1: The EMODnet marine data service Portal (infographic, 2023, ©EMODnet Secretariat)

Finally, KL noted that in addition to funded partners, EMODnet also has an Associated Partnership scheme that is a flexible, no-fee network for stakeholders, including the private sector, who would like a closer dialogue with EMODnet.

Participants were invited to take part in two online polls with questions related to the geographic scale of their activities and their role(s) across the marine knowledge value chain (see poll results in the Annex). Most respondents indicated that their NGOs are active in ocean conservation and monitoring, marine litter reduction, advocacy and awareness, and ocean restoration. In terms of scale, the majority operate at the European level, followed by regional sea-basin, with some NGOs operating internationally and others focusing more at a local level.

Demonstrating the EMODnet *in situ* marine data 'offer' for NGOs

The first workshop session featured demonstrations from various EMODnet Coordinators and experts from the EMODnet Geology, Physics, Seabed Habitats, and Human Activities thematics who showcased the diverse EMODnet *in situ* marine data offer in the context of NGO activities. This included providing examples of how NGOs can use these open data resources to support environmental monitoring, advocacy, Maritime Spatial Planning, and wider applications.

EMODnet Geology offer

Anu Kaskela (AK, GTK, Finland, EMODnet Geology Coordinator) presented EMODnet Geology, which is delivered by a network of 40 organisations across 30 countries. Her presentation covered key data themes such as deep-sea geology, coastal erosion, and geohazards, highlighting their relevance for Maritime Spatial Planning (MSP), resource management, and public awareness. AK also shared a

practical example of an EMODnet Associated Partner using EMODnet Geological data to support technical studies on offshore energy potential in Portugal.[5]

EMODnet Physics offer

Valeria Pizziol (VP, ETT, Italy) presented the EMODnet Physics data offer, highlighting its diversity, which includes temperature, salinity and metocean parameters, also extending to underwater noise, sea level measurements and more. She highlighted the continuous data flow into EMODnet, which includes delayed mode and near real-time data, and the development of a citizen science pipeline that integrates data from sailing and scuba diving computers. Ocean temperature products, climatology maps, and sea level anomaly visualisations were showcased as valuable tools for NGOs, noting that EMODnet offers many physical parameters at a global scale.

EMODnet Seabed Habitats offer

Ville Karvinen (VK, Syke, Finland, EMODnet Seabed Habitats Coordinator) introduced seabed habitat survey data and composite products relevant for NGOs, including Essential Ocean Variables (EOVs), Essential Fish Habitats (EFHs), Vulnerable Marine Ecosystems (VMEs), and Regional Sea Convention-specific datasets. He also presented the EMODnet data product EUSeaMap, which is a unique EU asset, providing a broad-scale modelled seabed habitat map that integrates multiple EMODnet layers to support MSFD benthic habitat types and EUNIS classifications. Practical use cases included the use of the EUSeaMap by the UK Blue Carbon Inventory (2024), and it was noted that EMODnet Seabed Habitats also provides expert advice to European initiatives and policymakers on marine data and knowledge related to Blue Carbon and to support implementation of the EU Nature Restoration Law.

A question was raised about the under-representation of Baltic Sea data, particularly seagrass in Sweden. VK acknowledged the existing data gap and confirmed that updates that were scheduled for October 2025 would include data from the HELCOM Biodiversity Database, encouraging NGOs to contribute new datasets for publication.

EMODnet Human Activities offer

Alessandro Pititto (AP, COGEA-Bip Group, Italy, EMODnet Human Activities Coordinator) concluded the session with a live demonstration of the EMODnet Common Map Viewer, focusing on Human Activities data layers. These include Marine Protected Areas (MPAs) such as Natura 2000 and the World Database on Protected Areas, harmonised MSP layers, filter options for relevant activities like ocean energy facilities, and the vessel density data product.

KL thanked all presenters, noting there are seven EMODnet thematics which also include marine chemistry, biodiversity and bathymetry. She invited NGOs and all participants to look at the EMODnet Portal to discover the full offer and see how EMODnet could support their activities.

1.2 Use Cases: The power of EMODnet's actionable marine data for NGOs

Session one continued with a showcase of use cases on how open-source EMODnet data and data products are already supporting the NGO sector for their ocean and coastal-related activities. Three NGO representatives shared practical examples of how their organisations currently use EMODnet services, as well as emerging opportunities to strengthen marine conservation and advocacy efforts. The discussion also explored ways to promote EMODnet across wider networks and gather NGO input on data gaps and future product needs.

[5] https://emodnet.ec.europa.eu/en/use-case/emodnet-data-supporting-technical-studies-offshore-energy-potential-portugal

Venice Lagoon Plastic Free

Davide Poletto (DP, Venice Lagoon Plastic Free, Italy) presented his organisation's work using the Venice Lagoon—a UNESCO World Heritage site—as a living laboratory for marine litter prevention and monitoring. He introduced their digital beach litter monitoring application (app), developed under the European Remedies Project, noting that the app integrates EMODnet Chemistry marine litter data. This tool supports citizen science initiatives and enables scalable monitoring strategies. DP emphasised how digital solutions powered by EMODnet data enhance collaboration and amplify local environmental impact.

WWF European Office

Laura Gillevic (LG, WWF European Office, Belgium) explained how WWF leverages EMODnet services to centralise diverse marine datasets, which are critical for producing evidence-based policy materials. She highlighted the importance of harmonised, open data for spatial analyses related to seabed habitats, maritime activities, and fisheries. LG also expressed interest in expanding WWF's use of EMODnet in line with evolving EU policy frameworks, such as the European Ocean Pact, the revision of the Marine Strategy Framework Directive (MSFD), and the proposed Ocean Act. She suggested future developments could include near real-time data on MPAs and integration with National Restoration Plans in 2026.

KL noted that while EMODnet does not conduct assessments, it provides a foundational service for such work, and NGO contributions to MSFD working groups were encouraged to support data standardisation and accessibility.

Oceana

Jorge Blanco (JB, Oceana, Portugal) described how Oceana uses EMODnet's data offer to support ocean protection. EMODnet data underpins Oceana's workflow, from initial analyses to oceanographic expeditions and final outputs. For instance, EMODnet Bathymetry data informs fishing activity assessments, EMODnet Seabed Habitat data supports impact analysis and management proposals, and EMODnet Chemistry data validates plastic dispersion models and pollution levels. JB also noted that EMODnet Human Activities and Physics data are used by Oceana in climate change studies and conservation planning and highlighted Oceana's commitment to open data, noting that the organisation shares its own datasets via the EMODnet Ingestion Portal, ensuring accessibility and long-term preservation. Oceana also uses EMODnet to provide the foundational data used to select expedition survey locations and is frequently featured in Oceana's reports, scientific publications, and policy responses. Its comprehensive coverage of European seas makes EMODnet an indispensable resource for Oceana's mission.

Q&A

The Q&A session addressed a range of questions raised both during the live discussion and via the online chat. One participant suggested that EMODnet could serve as a central portal for all relevant data from publicly funded projects, noting that when project websites become inactive, valuable data is often lost. VK explained that many EU Horizon programme calls now require marine and coastal *in situ* data submission to EMODnet and he gave an example that the EMODnet Products Catalogue[6] already hosts eight datasets related to the Mediterranean Sensitive Habitats (MediSeh) project.[7] The Secretariat confirmed that EMODnet also provides guidance to European projects on preparing their

[6] https://emodnet.ec.europa.eu/geonetwork/srv/eng/catalog.search#/search?from=1&to=30
[7] https://emodnet.ec.europa.eu/geonetwork/srv/eng/catalog.search#/metadata/6d617269-6e65-696e-666f-000000005103

data and metadata for submission to EMODnet and that a guidance document and recording of an EMODnet webinar for European projects are publicly available on the EMODnet Portal.[8] Participants were invited to contact the EMODnet helpdesk (helpdesk[at]emodnet.ec.europa.eu) to share further information on potential datasets that could be relevant for ingestion into EMODnet.

Some thematic-specific questions included the EMODnet offer of seabed habitat data in the Baltic, whether EMODnet offers sewage discharge information, to which AP responded that EMODnet Human Activities includes treatment plant and discharge point layers under the Urban Wastewater Directive, updated annually by the European Environmental Agency (EEA).

Participants also asked about citizen science data, noting that some NGOs work with citizen science initiatives, e.g., for collecting (marine/coastal) biodiversity data, among others. KL explained that EMODnet offers a central hub for *in situ* citizen science data related to the ocean and coast (and where possible inland waters), and many thematics are already integrating citizen science data into their composite data layers and data products.

Following a question on how EMODnet collects user metrics, KL clarified that EMODnet monitors overall usage through website metrics in terms of the number of unique visitors, data downloads, etc. It cannot track individual users as EMODnet is an official EC public marine data service on the Europa domain.

1.3 Discussion Session 1: EMODnet – NGO use cases and applications for existing and emerging marine data needs and requirements by the ocean-NGO sector

The first discussion session focused on the use of EMODnet by NGOs and explored applications for both existing and emerging marine data needs within the ocean-NGO sector. The conversation was guided by a series of structured questions, each followed by stakeholder feedback, providing insights into current practices, challenges, and opportunities for strengthening collaboration and data accessibility.

What types of Marine environment and human activities at sea data do you need, and why?

Participants discussed the types of marine environmental and human activities data most needed by NGOs and the reasons behind these requirements. Key priorities included habitat and species data for the Black Sea, live vessel location information to reduce collision risks with autonomous underwater vehicles, and harmonised datasets across EU countries covering MSPs, Marine Protected Areas (MPAs), and marine habitats. Questions were raised about the source of historic vessel density data and the availability of near real-time Automatic Identification System (AIS) streams, which AP clarified are based on AIS data purchased from commercial providers and processed annually, with efforts to include more recent data. NGOs also expressed interest in fisheries activity and fishing effort data similar to Global Fishing Watch, ideally in real time. AP explained that EMODnet Human Activities provides two relevant layers—one from the EC Joint Research Centre (JRC) covering the EU and another from the International Council for the Exploration of the Sea (ICES) for ICES areas.

Additional requests included underwater noise data, which EMODnet Physics confirmed is available and being expanded, and Vessel Monitoring System (VMS) data for fisheries, which AP noted is highly sensitive and there were challenges in making these data public.

[8] https://emodnet.ec.europa.eu/en/emodnet-guidelines-european-project-data-submissions

Participants agreed that all geographical scales of data were important, and they highlighted the value of *in situ* data for regional sea-basin, National and local applications, including for ocean protection (e.g., MPA) applications. Data at all levels would be beneficial for their work.

KL ended the discussion session by emphasising that EMODnet's role is to provide trusted, authoritative pan-European data layers rather than navigation-grade or real-time services, although some near real-time capabilities exist for specific thematics.

How can EMODnet serve your emerging marine knowledge requirements?

When asked how EMODnet could support emerging marine knowledge requirements, it was suggested to have a more regular process for bringing in user requirements of NGOs and of National administrations for EU Directives, for example, to provide fit-for-use data to assist Member States in implementing key measures, such as the identification and designation of Special Protection Zones (SPZ) and Natura 2000 sites.

What are the Priority gaps to address from the ocean-NGO perspective?

Participants identified several critical areas. Pollution in its various forms and climate change were highlighted as top concerns, with recognition that EMODnet's Chemistry offer already provides extensive data on contaminants, heavy metals, and pollutants, though further improvements in coverage and resolution are needed. Biodiversity loss was another key issue. NGOs recognised EMODnet's Biology offer, which includes occurrence and distribution data for marine species, and noted that NGOs would like to have additional datasets, e.g., more parameters and higher temporal and spatial resolution to better understand trends and impacts. The Blue Economy was also mentioned, with participants directed to the EMODnet's Human Activities offer, available via the EMODnet Map Viewer and catalogue. The Secretariat also noted that EMODnet is continuously optimising its metadata and is looking at metadata tagging, e.g., of data relevant for EU Directives, to enable easier data discovery.

A question was raised on how EMODnet integrates marine environmental and human activities data from EU Member States that are collected for legislative monitoring purposes, e.g., for the EU Marine Strategy Framework Directive (MSFD) and the Maritime Spatial Planning Directive (MSPD). KL noted that AP had presented that EMODnet Human Activities offers EU Member State National Maritime Spatial Plans (MSPs) on EMODnet, working with Member States to convert their MSP into geospatial data layers that are harmonised across regional sea-basins, enabling transboundary co-operation and information sharing. For MSFD, she confirmed that EMODnet experts are providing advice on data and metadata standards and data flows to MSFD Working Groups and the Technical Group on Data. Currently, EMODnet does not comprehensively integrate these data, but the EMODnet partnership is working closely with EC DG MARE, DG ENV, EEA and is increasing its engagement with Member States to achieve this in the coming years.

A strong call was made for harmonised data collection and publication across all sea basins and Member States to support the effective implementation of EU marine legislation. Specific gaps related to existing and emerging EU policy include more holistic habitat (seabed and water column) data for nature restoration plans, more comprehensive data related to all MSFD descriptors, EU Member State plans, noting MSP is ongoing and that EMODnet could also host Nature Restoration Plans, and regularly updated information on MPAs.

KL noted that EMODnet is continuously evolving its services and that the European Ocean Pact, EU Ocean Observation initiative and European Digital Twin Ocean will require a step-change in

EMODnet's provision of actionable *in situ* marine and coastal data. She emphasised that the EMODnet partnership has the expertise to do so, yet requires appropriate and strengthened investment to be able to accelerate and scale data sharing and integration. She referred participants to the EMODnet community Vision 2035, which would be launched in November 2025.[9]

Session 2: NGO data sharing to enhance the marine knowledge base

2.1 EMODnet – NGO data sharing successes

KL opened the second session, noting that this focused on NGO data sharing. The session showcased EMODnet data sharing success stories[10] with concrete examples of NGOs sharing ocean data, presented by NGO representatives and by EMODnet thematic experts.

NGO marine, coastal and beach litter

Matteo Vinci (MV, OGS, Italy) presented Success Stories of multiple NGO data sharing with EMODnet Chemistry, using marine litter data as an example, including beach litter, seafloor litter, and micro-litter. He explained the importance of EMODnet's expert network that develops and uses best-practice formats, rigorous quality control, and harmonisation, as these underpin EMODnet data products. He also referred to the use of EMODnet's marine litter database in policy reports, such as the JRC Technical Report on Marine Litter[11] and the European Coastline Macro Litter Trends[12] reports. MV highlighted collaborations with NGOs like MarineLitterWatch and Keep Sweden Tidy, whose citizen science data is officially recognised for MSFD Descriptor 10 evaluations in Sweden. He also outlined EMODnet Chemistry's role in linking EU-level efforts with global initiatives, including UNEP's Global Partnership on Marine Litter and G20 projects.

Marine Litter in the Black Sea – A Growing Challenge, Mare Nostrum Experience

Marian Paiu (MP, Mare Nostrum, Romania) described Mare Nostrum's long-standing collaboration with EMODnet Chemistry, focusing on citizen science initiatives for marine litter monitoring and ocean literacy. He detailed methodologies for beach litter monitoring, results from recent projects such as BlackNETs, which focuses on the removal of ghost nets, and the integration of these datasets into EMODnet Chemistry to support Good Environmental Status (GES) under MSFD Descriptor 10.

Harnessing the power of NGOs as Marine Biodiversity Data Providers

Cyril Radermecker (CR, VLIZ, Belgium) provided an overview of EMODnet Biology's data workflow, emphasising how NGO data submitted to EMODnet is integrated into EMODnet data layers and data products, with the full offer made available to global data initiatives such as the Ocean Biodiversity Information System (OBIS) and Global Biodiversity Information Facility (GBIF), and also feeding into the European Digital Twin of the Ocean. CR showcased three NGO examples: Mare Nostrum (Romania), whose CENOBS project data supports MSFD implementation in the Black Sea; Marecamp (Italy), which collects cetacean observations via citizen science; and Nereide (Spain), which shares biodiversity monitoring data through EMODnet Biology.

Near real-time glider data to EMODnet: Voice of the Ocean

Callum Rollo (CR, VOTO, Sweden, EMODnet Associated Partner) presented VOTO's near real-time data sharing approach, using underwater gliders to collect physical and biogeochemical data in the Baltic Sea. He explained the data pathway from *in situ* measurements to EMODnet via ERDDAP, enabling

[9] https://emodnet.ec.europa.eu/en/emodnet-community-vision-2035
[10] https://emodnet.ec.europa.eu/en/data-sharing-success-stories
[11] https://publications.jrc.ec.europa.eu/repository/bitstream/JRC133594/JRC133594_01.pdf
[12] https://publications.jrc.ec.europa.eu/repository/bitstream/JRC138907/JRC138907_01.pdf

rapid access through the EMODnet Common Map Viewer. CR also demonstrated how VOTO combines EMODnet and Copernicus Marine Service[13] data to identify ocean mixing fronts for field surveys. KL highlighted the growing opportunities for autonomous and near real-time data integration across EMODnet thematics.

OutBe Citizen Science Physical Data into EMODnet

Finally, Arianna Liconti (AL, OutBe, Italy) introduced OutBe's citizen science model, where coastal communities collect physical data during water sports activities using low-tech instruments. She described OutBe's collaboration with EMODnet to ensure these datasets are accessible and reusable, including their contribution to large-scale initiatives such as the Ocean Race, where all participating vessels collected physical data for EMODnet.

KL thanked all speakers, noting that these were excellent examples of how NGOs already contribute to EMODnet and that there are many more, encouraging NGOs that have not yet shared data to contact EMODnet with initial lists of available datasets so that EMODnet experts can facilitate integration.

2.2 EMODnet Guidelines for Data Sharing and Integration

Dick Schaap (DS, MARIS, The Netherlands, EMODnet Data Ingestion Coordinator) provided an overview of the EMODnet Data Ingestion service, which enables marine data holders to share their datasets for integration into EMODnet. He described this service as a key component of EMODnet's public offering, designed to democratise access to marine data and particularly aimed at sectors such as NGOs and citizen science initiatives that often lack established data pipelines. DS outlined the ingestion process and the Marine Data Management Guidelines[14] for data and metadata, emphasising the importance of compliance and quality standards. He highlighted successful examples of NGO collaboration, including Mare Nostrum and the Keep Sweden Tidy Foundation, which share marine litter data through EMODnet Ingestion with strong adherence to best practices. DS also noted that these partnerships have led to structured, long-term data flows that are win-win.

2.3 Discussion Session 2: Data sharing with EMODnet: Towards Marine Knowledge for Society

KL opened the floor for the second discussion session, focused on promoting NGO-ocean data sharing with EMODnet to strengthen marine knowledge for society. Participants explored the mutual benefits of collaboration and provided feedback on existing and emerging data needs of the NGO sector.

Is your NGO collecting in situ ocean observations? Are these data shared with EMODnet?

Several NGOs confirmed they collect relevant data, with some data sharing with EMODnet, including VOTO, Oceana, and all NGOs that had presented data sharing success stories. EMODnet representatives provided more information on data flows to EMODnet. They explained that many public research organisations already have established pipelines via national oceanographic data centres. The EMODnet Data Ingestion service focuses on connecting NGOs, citizen science initiatives, and private actors that lack such infrastructure, capacity, so that data can be ingested and integrated, and more operational data flows can then be established. He emphasised that EMODnet's Data Ingestion is not an *ad hoc* process, but a structured approach designed to ensure compliance with standards and create long-term, high-quality data flows. Support is available for new providers during the early stages, including guidance on formatting and quality checks, with the intention to build lasting collaborations. He encouraged participants to consult EMODnet's data ingestion guidelines

[13] https://marine.copernicus.eu/
[14] https://emodnet.ec.europa.eu/en/data-ingestion#marine-data-management-guidelines

(see footnote 16) and explore dedicated services, such as those for beach litter monitoring, available on the EMODnet portal.[15]

What could be the next steps to enable more NGOs to share data?

Participants raised questions about the integration of existing datasets, such as marine mammal data from OBIS Seamap, to avoid duplication of effort. EMODnet representatives confirmed that OBIS Seamap data is already incorporated into EMODnet, with ongoing work to strengthen this integration. WWF offered to connect EMODnet with colleagues managing the Blue Corridors project, which compiles whale migration data, noting its potential relevance for EMODnet Biology. CR confirmed that these datasets are not yet present in EMODnet's system and agreed that they would be valuable additions.

How could EMODnet scale up its connections with NGOs to increase and accelerate data sharing? Are there any challenges or barriers to data sharing? How can these be overcome?

Participants discussed how EMODnet could strengthen its connections with NGOs to accelerate data sharing. Suggestions included joint projects to promote successful examples of collaboration and creating opportunities for NGOs to showcase best practices. One participant highlighted an initiative by Scuba Network, which connects divers with ocean NGOs to share dive logs and observations of marine life and debris, noting that broader outreach support would help scale this effort. The discussion also proposed organising dedicated events or expanding existing EMODnet activities to bring NGOs together on a larger scale and foster cross-network collaboration.

Participants discussed the challenges and barriers to data sharing and how these could be addressed. Time constraints were identified as a major issue, with NGOs noting that limited staff capacity often makes it difficult to allocate resources for data sharing. A lack of awareness about the value of sharing data and the steps involved was also highlighted. To overcome these barriers, participants suggested clearer guidance for new contributors and easier access to instructions on the EMODnet portal. EMODnet representatives pointed to existing resources, including the Data Ingestion Portal[16] and a free self-paced online course for EMODnet Biology[17], as well as technical reports and videos explaining the process behind data product creation. KL acknowledged that sensitivities around commercial data are common, but encouraged NGOs to explore available tools and guidelines[18] to streamline data submission and ensure compliance with standards. An additional comment was raised regarding macroplastic monitoring, specifically whether it is possible to access logs showing the most common items found at monitored sites. In response, MV explained that EMODnet Chemistry provides products on its portal that include the locations of newly surveyed beaches and relevant data, which could help answer this question within the scope of existing products.

A word cloud (Annex) was conducted on what applications the participants would use EMODnet for in the future, with examples including MPAs and the 30x30, Climate Change, the Ocean Act and near real-time data as some examples.

Key Outcomes and Next Steps

The workshop reaffirmed the critical role of NGOs as key stakeholders in ocean governance, policy advocacy, research, and innovation. NGOs are not only end-users of marine data but also important

[15] EMODnetEUprojectDataSubmissionGuidelines_March2024.pdf
[16] https://emodnet.ec.europa.eu/en/data-ingestion
[17] https://classroom.oceanteacher.org/course/view.php?id=958
[18] https://emodnet.ec.europa.eu/en/node/50317

producers of ocean observations through projects and citizen science initiatives. Their contributions are essential for evidence-based assessments and for advancing the objectives of EU marine policies. Discussions highlighted a strong interest in expanding and systematising NGO data contributions to EMODnet. Participants recognised that integrating NGO datasets more effectively would reduce uncertainty, strengthen decision-making, and enhance the collective marine knowledge base. Several concrete examples of successful NGO-EMODnet collaborations were presented during the workshop, demonstrating the value of structured partnerships. However, there was consensus that these efforts need to be scaled up and made more systematic to ensure comprehensive, high-quality marine data coverage across all European sea basins and beyond.

Looking ahead, EMODnet will continue to engage NGOs through targeted outreach and collaborative initiatives and to explore ways to improve and expand data sharing and integration from NGOs into EMODnet, whilst also maximising NGOs' awareness of the value and benefits of EMODnet to support their activities. Participants were encouraged to provide feedback on EMODnet services, share datasets, and propose new use cases that could be featured on the EMODnet Portal. EMODnet will also work to align its services with emerging policy priorities, including the European Ocean Pact and nature restoration objectives, ensuring that NGOs have access to harmonised, authoritative data to support their advocacy and conservation work. The EMODnet Open Conference 2025 featured a dedicated breakout session on “Data Sharing and Integration with EMODnet,” providing a key platform to advance these discussions and scale up NGO involvement. By building on these outcomes, EMODnet aims to create a more inclusive and collaborative marine data ecosystem, ensuring that NGOs can both contribute to and benefit from Europe’s most comprehensive marine knowledge base.

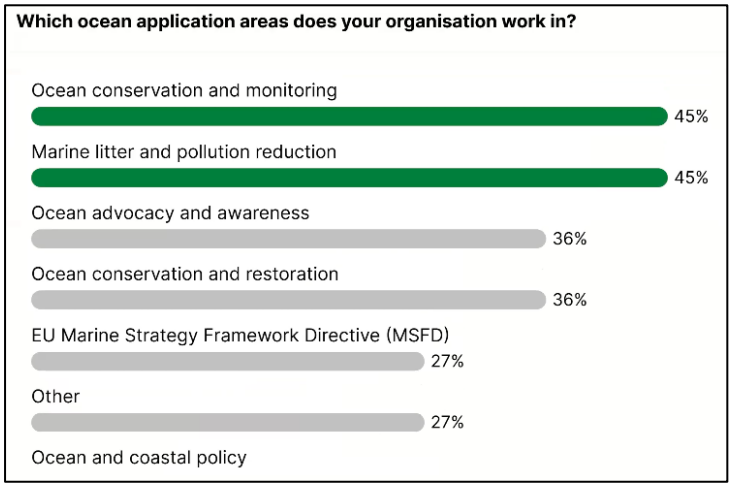
Closing Remarks from EMODnet

KL closed the workshop by thanking the co-organisers and all participants, noting that EMODnet welcomes further dialogue with NGOs in the future. She confirmed that a full summary report (this document) would be published on the EMODnet Portal and the European Commission’s Maritime Forum[19], alongside reports from previous EMODnet workshops which included EMODnet for coastal tourism, offshore renewable energy, marinas, recreational boating and ports and aquaculture. Participants were invited to maintain dialogue with EMODnet and its stakeholder networks to maximise the benefits of EMODnet for NGO activities, enhance data sharing, and ensure the availability of comprehensive marine knowledge to support evidence-based policy making. KL encouraged NGOs to contact EMODnet (via the helpdesk or to the Secretariat) for collaborative use case development and to share data sharing success stories. She also highlighted opportunities for engagement through the EMODnet Open Conference on 25-26 November 2025 and the EMODnet Associated Partnership Scheme.[20]

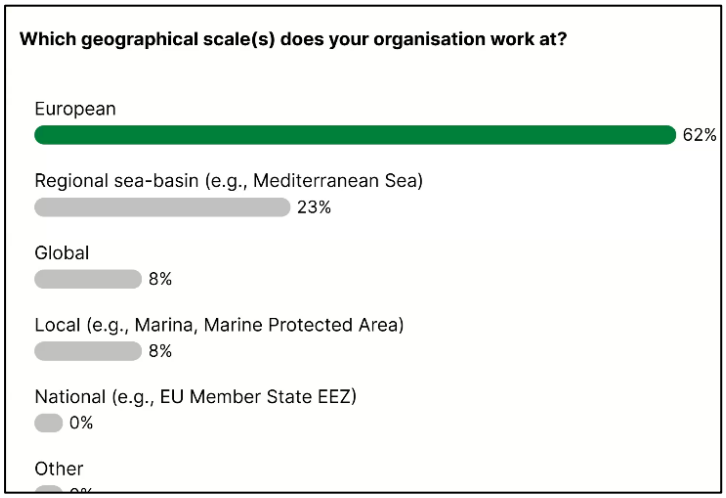
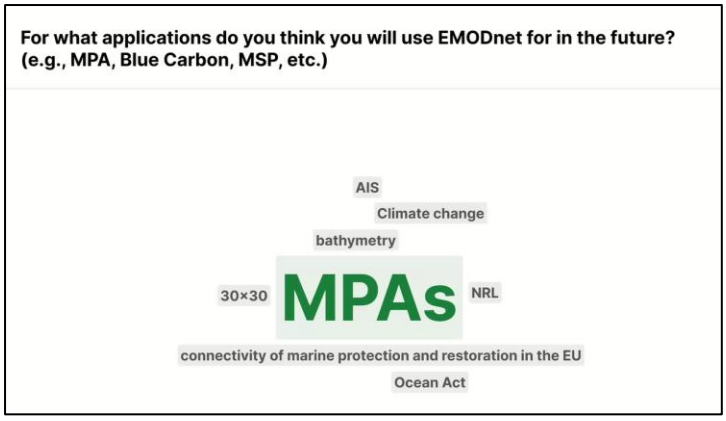
[19] https://maritime-forum.ec.europa.eu/index_en
[20] https://emodnet.ec.europa.eu/en/emodnet-associated-partnership-scheme

Annex: Polling questions

Poll 1.



Poll 2.



Word Cloud 1.